

TABLE I
SIMULATION PARAMETERS

Parameter	Value / Description
Number of smart bins, N	84
Scalability values of N	84, 100, ..., 500
Baseline road network	OpenStreetMap/SUMO
Time-dependent traffic	Hourly SUMO/TOPIS travel times
Number of vehicles, K	10, $\{6, \dots, 14\}$ for sweep
Scalability fleet size	approximately $N/8.4$
Curb mass, m_0	7,000 kg
Payload capacity, Q	2,000 kg
Maximum battery capacity, $B_{\max}$	180 kWh
Minimum battery reserve, $B_{\min}$	15 kWh
Node weight, w_i	$\mathcal{U}(50, 250)$ kg
Rolling coefficient, C_{rr}	0.010
Drag coefficient, C_d	0.60
Frontal area, A_f	5.5 m ²
Air density, ρ	1.225 kg/m ³
Propulsion efficiency, η_{drv}	0.90
Regenerative efficiency, η_{reg}	0.60
En-route auxiliary power, P_{aux}	3 kW
Stop-and-go threshold, ΔT	$0.25T^{ff}$
Stop-and-go coefficient, κ_{sg}	4.93×10^{-6} kWh/kg
Service time per bin, T_s	120 s/bin
Stationary auxiliary power, P_{svc}	2 kW
Lifting/compaction energy, e_i^{lc}	0.12 kWh/bin
Payload state interval, $\Delta \ell$	100 kg

I. PERFORMANCE EVALUATION

This section evaluates the proposed cooperative assignment-and-routing framework in an electric waste-collection digital twin constructed from a real urban road network. The experiments address four questions: (i) whether the proposed method consistently reduces operating energy under the hourly traffic conditions of the service area, (ii) how the exact number of active vehicles changes energy consumption and completion time, (iii) whether replanning is beneficial when localized traffic disruptions evolve during operation, and (iv) whether the solution quality is retained as the number of service bins increases.

A. Simulation setup

The evaluation uses the Seongbuk-gu road network in Seoul. The network is generated from OpenStreetMap, represented as a directed SUMO graph, and augmented with road grade and hourly link travel time. The base instance contains $N = 84$ measured waste-bin locations. For every transition, all methods use the same load-, grade-, speed-, and time-dependent energy model in Section II, including auxiliary, stop-and-go, service, and lifting/compaction energy. Therefore, the reported energy is battery energy rather than distance or a surrogate routing cost.

Unless otherwise stated, the base experiments use $K = 10$ vehicles and demand seed 7. Every returned plan must service every bin exactly once, respect the 2,000 kg vehicle capacity and 15 kWh battery reserve, return every vehicle to the depot, and assign at least one bin to each of the exactly K active vehicles. Infeasible solutions are not assigned a finite objective value.

The proposed method is compared with a state-aware nearest-neighbor method (NN), ant-colony optimization (ACO), particle-swarm optimization (PSO), and a genetic algorithm (GA). ACO uses 60 ants and 150 iterations, PSO uses 60 particles and 200 iterations, and GA uses a population of 100 and 500 generations. Their candidate permutations are decoded into exactly K nonempty, capacity- and battery-feasible routes by the same dynamic-programming split procedure. Thus, the comparison changes the search method while keeping the physical evaluator and feasibility rules fixed. The MILP implementation is retained for small-instance correctness tests, but is not used as an operational-size baseline. If the loopy message updates do not satisfy the strict residual test within the prescribed round budget, the best feasible incumbent is reported; consequently, the experiments do not claim global optimality.

B. Hourly traffic and route comparison

The first experiment fixes the bin demands and evaluates all methods for the 24 traffic snapshots from 00:00 to 23:00. Figure 1 shows the routes obtained at 11:00 by the proposed method and the two strongest stochastic baselines in this snapshot. The proposed method uses 7.547 kWh less than ACO and 23.843 kWh less than GA. Its 3,261.6 s makespan is nevertheless longer than the 2,809.1 s and 2,806.8 s obtained by ACO and GA, respectively. This example illustrates that the optimized objective is fleet energy, not makespan, and that the two metrics need not rank route sets identically.

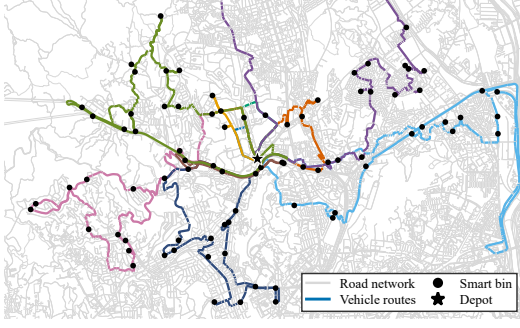
As shown in Fig. 2, the proposed method has the lowest energy in all 24 matched hourly comparisons. Its mean energy is 116.893 kWh, compared with 121.961 kWh for ACO, 130.857 kWh for GA, 141.951 kWh for PSO, and 148.240 kWh for NN. This corresponds to reductions of 4.16%, 10.67%, 17.65%, and 21.15%, respectively, when the ratio of the 24-hour means is used. Even the smallest hourly margin over ACO is 1.623 kWh.

The energy benefit is not obtained by uniformly lengthening vehicle operation. The mean active-vehicle completion time of the proposed method is 1,631.8 s, lower than ACO (1,679.2 s), GA (1,743.5 s), PSO (1,817.2 s), and NN (1,864.0 s). The proposed energy ranges from 78.198 kWh at 04:00 to 168.065 kWh at 12:00. The nonmonotonic hourly pattern results from the spatial distribution of link-specific congestion and the stop-and-go term; a single network-wide average speed is therefore insufficient to predict the energy of the selected routes.

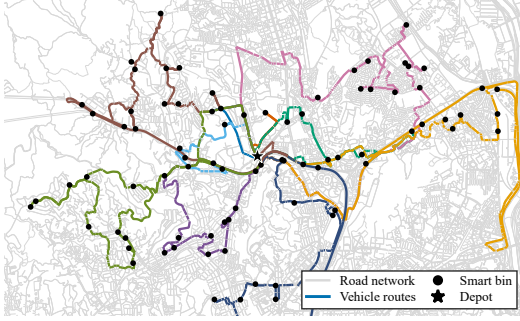
C. Effect of the exact number of active vehicles

The second experiment varies the exact number of active vehicles from 6 to 14. The total demand is fixed at 11,700 kg, so six vehicles provide only 300 kg of aggregate spare capacity, whereas larger fleets introduce additional depot-to-service and return legs. For a metric z , Fig. 3 reports

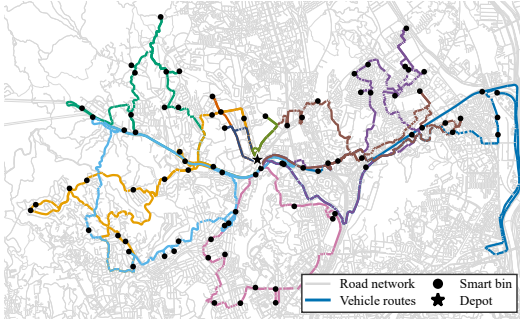
$$g_{m,K,h} = 100 \left(\frac{z_{m,K,h}}{\min_{m',K'} z_{m',K',h}} - 1 \right), \quad (1)$$



(a) Proposed: 164.679 kWh



(b) ACO: 172.225 kWh



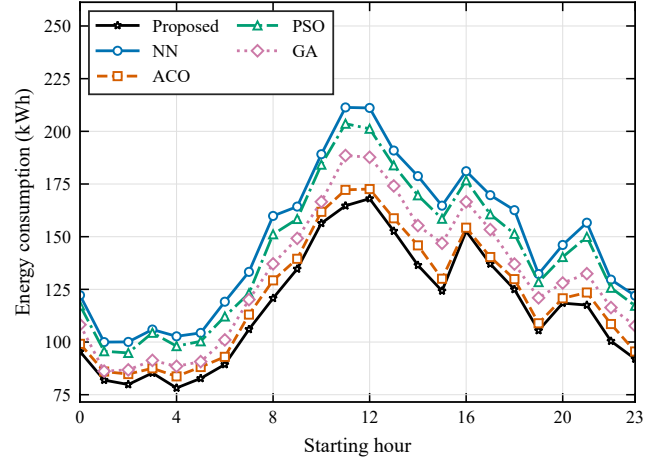
(c) GA: 188.522 kWh

Fig. 1. Fleet-route comparison on the same Seongsu-gu instance at 11:00. All methods service the same 84 bins with exactly 10 nonempty and feasible vehicle routes. Colors distinguish vehicle routes, black circles indicate smart bins, and the star indicates the depot.

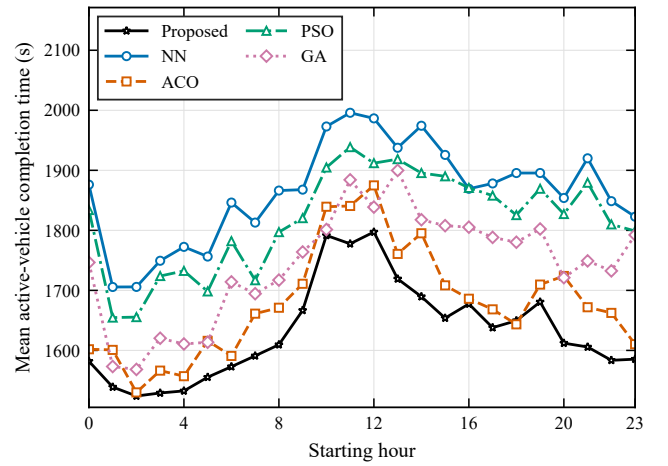
followed by the mean and range of $g_{m,K,h}$ over the three evaluated hours.

The proposed method consumes less energy than every baseline in all 27 matched hour-and- K cases. Averaged over these cases, it reduces energy by 5.55% relative to ACO, 10.62% relative to GA, 18.65% relative to PSO, and 22.30% relative to NN. Its three-hour mean energy is minimized at $K = 6$ (107.100 kWh); the individual-hour minima occur at $K = 7$ for 04:00 and at $K = 6$ for 11:00 and 19:00. Energy then generally increases as more vehicles repeat fixed depot-access and return overhead.

The completion-time result exhibits a different optimum. The proposed method's lowest three-hour mean makespan is 2,940.0 s at $K = 9$, rather than at its energy-optimal $K = 6$. Moreover, makespan is not monotone in K , because adding a vehicle changes both the assignment and the longest route. The



(a) Fleet energy consumption



(b) Mean active-vehicle completion time

Fig. 2. Performance over the 24 hourly traffic snapshots provided by [?]. The demand realization and fleet size are fixed across all hours and methods.

experiment therefore identifies an operational tradeoff instead of imposing a convex fleet-size curve: a smaller feasible fleet minimizes energy in this instance, whereas a moderately larger fleet improves completion time.

D. Online adaptation to localized disruptions

The third experiment begins at 18:00 and uses ten paired traffic seeds. For each seed, approximately 5% of the bins are selected as fixed disruption centers. Traffic remains normal for 10 min; speeds on edges within a 500 m road-distance radius then decrease linearly from 100% to 35% over 10 min, remain at 35% for 10 min, and recover linearly over 25 min. Every method observes the same trace and may replan every 5 min.

Three policies are distinguished. Frozen static retains both the initial bin order and its precomputed edge paths. Live-navigation static retains the initial bin order but follows minimum-energy paths under the evolving traffic. Adaptive additionally reoptimizes the remaining bin assignments and orders. Vehicles continue moving while planning is performed, so planning time is not converted into stationary energy or operating delay; it is measured separately. An adaptive candidate is accepted only if exact evaluation under the currently

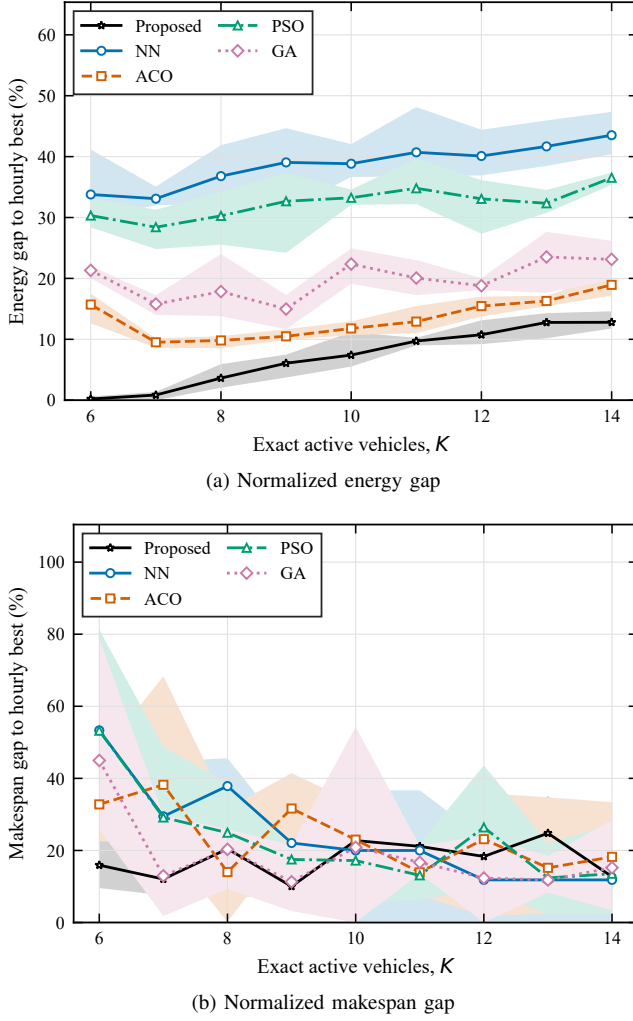


Fig. 3. Exact- K sweep at 04:00, 11:00, and 19:00 with $N = 84$ and 11,700 kg total demand. Each line is the three-hour mean gap to the best method-and- K value within the same hour, and each shaded region spans the minimum and maximum over the three hours.

observed traffic snapshot gives lower residual energy than the incumbent. No traffic forecast is supplied.

Figure 4 and Table II show that the proposed adaptive method has the lowest mean final energy, 158.313 kWh. It is 0.885 kWh below adaptive ACO and 21.725–38.597 kWh below adaptive PSO, GA, and NN. Relative to its own frozen-static execution, proposed replanning saves 10.555 kWh (6.04%) and wins in all ten paired seeds. Its cumulative post-departure replanning time is 9.76 min, lower than ACO, PSO, and GA but higher than NN.

Adaptive planning is not uniformly better than live-navigation static. For the proposed method, adaptive execution is 0.362 kWh higher on average and wins in six of ten seeds; GA shows a larger degradation. This does not contradict the acceptance test, which compares residual plans only under the current snapshot. A route accepted during degradation may lose its apparent advantage as traffic subsequently recovers. Thus, the experiment supports adaptive assignment under unanticipated path disruption, but also identifies the limitation of myopic replanning without adding a traffic-forecast model.

E. Scalability with the service-network size

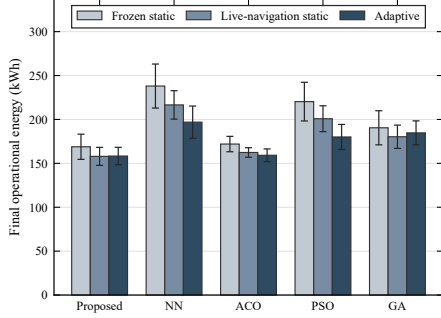
The final experiment uses nested instances $N \in \{84, 100, 120, \dots, 300, 350, 400, 450, 500\}$. The fleet size is adjusted to keep approximately 8.4 serviced bins per vehicle, so increasing N enlarges the number of coupled assignment edges while keeping the typical vehicle-level trellis size comparable. The demand seed changes both the waste realization and the randomized stochastic search, whereas the measured 84-bin locations remain fixed and the additional bins are drawn once from a seeded nested set of feasible SUMO road locations.

For the ten-seed portion currently complete through $N = 300$, every method returns feasible solutions. The proposed method has the lowest mean energy per bin at all 12 completed sizes. Its value changes from 1.963 ± 0.039 kWh/bin at $N = 84$ to 1.995 ± 0.042 kWh/bin at $N = 300$ and remains between 1.936 and 2.020 kWh/bin over the completed range. Averaged over the 12 sizes, the corresponding reduction is 2.85% relative to ACO, 10.73% relative to GA, 14.80% relative to PSO, and 16.80% relative to NN. The stable normalized energy indicates that total fleet energy grows mainly with the amount of serviced work rather than through a degradation of per-bin solution quality.

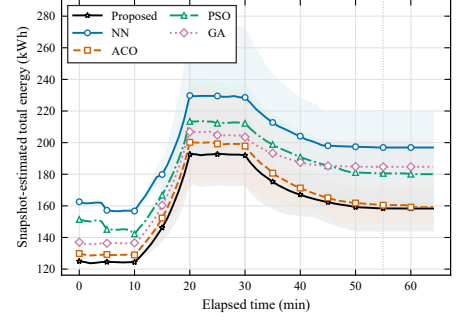
The makespan curve is less regular because it is determined by the single longest vehicle route and is not the optimized objective. For the proposed method, the mean makespan is 3,371.0 s at $N = 84$ and 4,284.2 s at $N = 300$, with intermediate changes caused by the road geometry, demand realization, and discrete fleet-size adjustment. Runtime is retained in the result files for reproducibility, but is not used as a primary figure because it combines hardware-dependent solver time with cost-oracle preparation; the analytical growth of the assignment and trellis updates is discussed separately in the complexity analysis.

F. Main findings

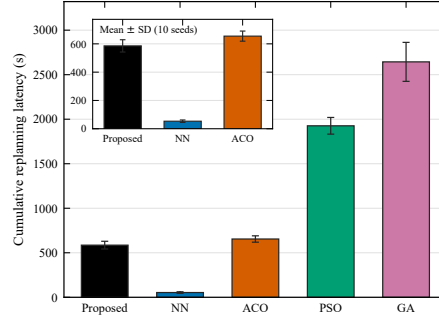
The experiments yield four main findings. First, coupling assignment messages with state-aware trellis feedback produces the lowest fleet energy across all 24 hourly snapshots and all 27 evaluated hour-and- K combinations, while maintaining exact coverage and feasibility. Second, fleet sizing is a multiobjective decision: the energy-optimal fleet contains six or seven vehicles in the evaluated range, whereas the proposed method's lowest mean makespan occurs at nine vehicles. Third, adaptive replanning is consistently beneficial relative to executing paths frozen before disruption, but it is not guaranteed to dominate live navigation when future traffic recovery is not modeled. Finally, the proposed energy per serviced bin remains stable and lowest among the evaluated methods as the network grows through the completed ten-seed range. Together, these results support the proposed architecture as an energy-oriented coordination method while clearly separating energy quality, completion time, online information, and computational scaling.



(a) Final operating energy



(b) Snapshot-estimated total energy

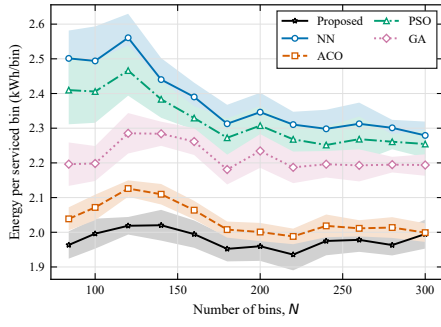


(c) Cumulative replanning latency

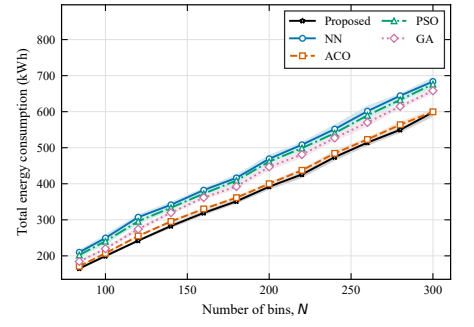
Fig. 4. Online performance under continuously developing and recovering localized disruptions. Lines and bars show means over ten paired traffic seeds; uncertainty intervals follow the definitions given in each figure.

TABLE II
ONLINE RESULTS OVER TEN PAIRED TRAFFIC SEEDS

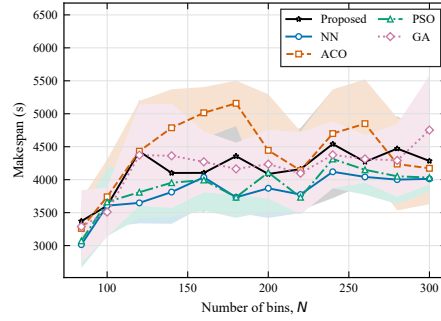
Method	Frozen static	Live static	Adaptive	Gain vs. live	Replanning latency
	Operational energy (kWh)			(%)	(min.)
Proposed	168.87 ± 14.33	157.95 ± 10.26	158.31 ± 9.92	−0.3	9.76 ± 0.72
NN	238.05 ± 25.05	216.54 ± 16.19	196.91 ± 18.35	9.0	0.89 ± 0.15
ACO	171.97 ± 8.76	162.38 ± 5.45	159.20 ± 7.24	1.9	10.91 ± 0.60
PSO	220.30 ± 22.02	200.82 ± 14.74	180.04 ± 14.27	10.2	32.10 ± 1.56
GA	190.50 ± 19.43	180.29 ± 13.23	184.70 ± 13.69	−2.5	44.07 ± 3.65



(a) Energy per serviced bin



(b) Total fleet energy



(c) Makespan

Fig. 5. Scalability with nested service-node sets and approximately 8.4 bins per vehicle. Lines show seed means and shaded regions show one sample standard deviation.